Nilaa Raghunathan

(943) 238-5010 | nr3005@columbia.edu | linkedin.com/in/nilaaraghunathan/ | github.com/Nilaa-Jaya

EDUCATION

Columbia University — **New York, US**
MS in Data Science — Sep 2024 - Dec 2025
- Coursework: Generative AI, Deep Learning, Machine Learning, Data Structures & Algorithms, Probability & Statistics.

Vellore Institute of Technology — **Vellore, IN**
BTech in Computer Science with Specialization in Data Science — Sep 2020 - May 2024
- Coursework: NLP, Artificial Intelligence, Business Intelligence, Cloud Computing, Data Analytics, Software Engineering.

TECHNICAL SKILLS

Languages: Python, R, SQL, C, C++, C#, Java, MATLAB.
Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, PySpark, Hugging Face Transformers, LangChain, FastAPI, PEFT.
Developer Tools: Docker, Kubernetes, CI/CD, AWS, Azure, Databricks, Apache Spark, Airflow, Kafka, PostgreSQL.

EXPERIENCE

Limex Trading — **New York, US**
Applied AI Intern — **Sep 2025 - Dec 2025**
- Reduced analyst research discovery time by 42% across 50 U.S. equities by building an agentic financial research copilot with Hybrid RAG, GraphRAG, and LangGraph-based multi-agent orchestration.
- Engineered Neo4j knowledge graph (35K+ nodes, 120K+ relationships) enabling multi-hop supply-chain & risk-exposure analysis.
- Improved retrieval precision by 24% and Recall@10 by 18% by implementing a hybrid retrieval pipeline combining vector search, BM25, Cypher graph traversal, Reciprocal Rank Fusion, and cross-encoder reranking.
- Fine-tuned Llama 3.1-8B with QLoRA (4-bit quantization) on 600+ GPT-4o-distilled financial instruction pairs, improving citation accuracy from 72% to 81% and answer faithfulness from 71% to 76%.

Magna International Inc. — **Michigan, US**
Data Science Intern — **Jun 2025 - Aug 2025**
- Improved an ML-based inventory demand forecasting system that decreased safety stock by 12% and saved $1.4M annually.
- Increased forecast accuracy by 18% (MAPE) using probabilistic time-series modeling (GMM-HMM) to capture dynamic demand patterns for supply planning.
- Strengthened demand signal reliability by building a context-aware NLP pipeline (BiLSTM-RoBERTa) over 7K+ supplier records.
- Lowered stock-out risk by 13% through ANOVA, A/B testing & causal impact analysis, validating changes before deployment.
- Delivered a Streamlit dashboard on Databricks with a MySQL backend for cross-functional teams.

Vellore Institute of Technology — **Vellore, IN**
Research Engineering Intern — **Apr 2022 - Jul 2023**
- Published first-author IEEE Access paper (80+ citations) on sentiment analysis, evaluating ML models across multilingual, cross-domain, and low-resource datasets.
- Architected a cloud-native NLP data platform on AWS, leveraging Kafka for streaming ingestion and Databricks/Spark for distributed processing of large-scale text datasets.
- Orchestrated Airflow DAGs for automated ETL, schema validation, data quality monitoring, and Spark-based feature engineering on Databricks, enabling scalable analytics and model evaluation across 50M+ text records.

PROJECTS

Generative Recommender System *(github.com/Nilaa-Jaya/Generative-Recommender-System)* — **Feb 2026 - Present**
- Designed RAG-based personalized-recommender system processing 230M+ reviews, 54M+ users & 1.57M+ products.
- Built scalable vector search system using Sentence-BERT, FAISS, diversity re-ranking, and persona-based retrieval via DeepMF embeddings (20 personas), improving Precision@10 from 0.57 to 0.69 and Diversity@10 from 0.42 to 0.51 with ~43ms latency.
- Fine-tuned LLaMA2-7B using QLoRA with RLHF reward model on 100K pairs, improving BERTScore from 0.79 to 0.86

Multi-Agent HR Intelligence Platform *(github.com/Nilaa-Jaya/Multi-Agent-HR-Intelligence-Platform-)* — **Dec 2025 - Feb 2026**
- Architected and deployed multi-agent HR support system using LangGraph/LangChain with 9 specialized agents, implementing 7-way conditional routing and stateful orchestration to automate 85% of HR inquiries with 90%+ resolution accuracy.
- Built scalable RAG infrastructure with FAISS vector search and production FastAPI microservices (<1s latency), PostgreSQL analytics, Docker deployment, GitHub Actions CI/CD (38 tests), and secure webhooks.

Financial Fraud Detection *(github.com/Nilaa-Jaya/Financial-Fraud-Detection)* — **Aug 2025 - Oct 2025**
- Developed fraud detection pipeline on 6.3M+ transactions, engineering error-based and anomaly features using Isolation Forest.
- Boosted fraud detection by ADASYN oversampling with ML (Logistic Regression, Random Forest, XGBoost) & deep learning models (RNN, LSTM, BiLSTM) achieving 0.98 F1-score, 97% recall & reducing false negatives by 50%.

PUBLICATIONS

- Challenges and Issues in Sentiment Analysis: Comprehensive Survey | *IEEE Access (Cited by 80+)*.
- Improved Deep Learning Model for Sentiment Analysis using LSTM with Layer Normalization | *IEEE - AAIAC Conference*.